Education

Shanghai Jiao Tong University, School of Computer Science, Electronic Information, Master's (recommended admission) **2026.9 – 2029.3**

Shanghai Jiao Tong University, School of Computer Science, Software Engineering, Bachelor's **2022.9 – 2026.6**

- **GPA 92.5/100, Rank 5/88, Undergraduate National Scholarship**

Project Experience

Innovator-VL: A Multimodal Large Language Model for Scientific Discovery **2025.10 – 2026.2**

(Project Lead & First Author) [GitHub] [Project Page]

- **Project Overview:** Led the development of a multimodal large language model designed for the scientific domain, open-sourcing **Innovator-VL-8B-Instruct** (general interaction) and **Innovator-VL-8B-Thinking** (deep reasoning), aiming to improve cross-disciplinary scientific understanding and reasoning.
- **Full-Pipeline Architecture:** Built a **fully transparent, end-to-end reproducible** training pipeline from data cleaning, pre-training and supervised fine-tuning (SFT) to reinforcement learning (RL). Optimized distributed training based on Megatron-LM, achieving efficient training throughput.
- **Extreme Data Efficiency:** Broke the traditional paradigm of relying on massive domain pre-training data. Through principled data filtering strategies, reached SOTA on multiple scientific and general vision benchmarks with only **fewer than 5 million** high-quality scientific samples.
- **RL Alignment (RLHF):** For the Thinking model, introduced the **GSPO** algorithm and a hierarchical reward mechanism, significantly strengthening multi-step reasoning (CoT); reasoning token efficiency improved **1.4× to 4.3×** over comparable models (e.g. Intern-S1-mini).
- **Open-Source Impact:** Gained **100+ GitHub Stars** and **2000+ HuggingFace downloads** shortly after release. Overall performance surpassed mainstream models such as Qwen3-VL-8B and InternVL3.5-8B across 37 benchmarks including Scientists' First Exam and MathVista.

AI for Service: Proactive Assistance with AI Glasses **2025.7 – 2025.10**

(Core Contributor & First Author) [arXiv]

- **Architecture Innovation:** Proposed the **Alpha-Service** framework and a "**Trigger-Analysis-Reasoning**" three-level multi-agent architecture; through modular perception–decision–memory design and large/small model cascade scheduling, achieved a balance between low latency and high performance on edge devices.
- **Deployment:** Built applications such as a game advisor and navigation guide on **Xiaomi AI glasses**, solving the "when to intervene" and "service strategy" problems of proactive service; the system accurately understands user intent and proactively offers assistance.

Research Experience

Activating Proactive Streaming Interaction in Video LLMs without Architectural Changes **2025.10 – 2026.2**

(First Author) Under Review [arXiv]

- **Background:** Addressing the core decision challenge that existing offline video LLMs (VideoLLMs) lack real-time interaction capability (i.e. not knowing "when" to speak), proposed a new paradigm that activates streaming video understanding **without modifying the model architecture**.
- **Streaming Evaluation Framework (RealStreamEval):** Built the first evaluation framework that models streaming video understanding as **frame-level multi-turn dialogue**; by introducing a strict **redundancy penalty mechanism**, it forces the model to decide response timing autonomously in real-time streams, solving traditional evaluation's dependence on external polling or offline segments.
- **Self-Evolving Data Synthesis (EvoStreaming):** Designed a **zero-human-annotation** self-evolving data generation pipeline; through a five-stage pipeline of "task classification – self-questioning – spatio-temporal anchoring – streaming trajectory synthesis – iterative refinement", uses the model's own capability to generate high-quality interaction signals. With only **1,000** self-generated samples, successfully activated streaming interaction in mainstream offline models such as Qwen2-VL and InternVL.
- **Performance Breakthrough:** Without sacrificing the offline model's general performance, achieved stronger streaming interaction using only **1/100 the data** of prior methods, improving by up to **10.8%** on OVO-Bench. This work reveals that existing video models already possess latent streaming understanding ability, and the core bottleneck is the lack of correct interaction-timing training signals.

Accelerating Streaming Video Understanding via Hierarchical Token Compression **2025.7 – 2025.11**

(Core Contributor) Under Review [arXiv]

- **Background:** Targeting the **compute bottleneck** of rapidly growing ViT encoding time in streaming video understanding, proposed a plug-and-play, **training-free** hierarchical acceleration framework (STC), breaking the limitation that traditional compression methods depend on future information or user instructions and achieving full-pipeline acceleration.
- **Core Algorithms:** Designed **STC-Cacher**, which uses a **Scatter-Update strategy** to compute only dynamic tokens and reuse the static background (avoiding 85% visual redundancy); proposed the **instruction-free STC-Pruner**, which precisely removes redundant features before the LLM based on dual-anchor spatio-temporal novelty.

- **Performance Breakthrough:** On four mainstream baselines including ReKV and Dispider, while maintaining **99% accuracy**, STC reduced ViT encoding latency by **24.5%** and LLM pre-fill latency by **45.3%**, significantly outperforming existing SOTA compression methods such as ToMe and VidCom².